ASSESSING MOUTH, THROAT, NOSE, & SINUSES

WEBER: CHAPTER 18

NURS3120: HEALTH ASSESSMENT AND PROMOTION

OBJECTIVES

- Describe the structure and function of the mouth, throat, nose, and sinuses
- Discuss risk factors for oral cancer and ways to reduce one's risks
- Differentiate between normal and abnormal findings of the mouth, throat, nose, and sinuses
- Analyze interview and physical assessment data related to the mouth, throat, nose, and sinuses to formulate valid clinical judgments

STRUCTURE & FUNCTION

Mouth:

- First part of the digestive tract; ; responsible for receiving food, tasting, preparing food for digestion & aiding in speech
- Cranial nerves V (trigeminal), VII (facial), IX (glossopharyngeal), & XII (hypoglossal) assist these functions
- Airway for the respiratory tract

Throat:

- Muscular passage for food & air
- Located behind the mouth & nose
- Made up of the nasopharynx (top), oropharynx (middle), laryngopharynx (bottom)
- Contains the tonsils; mounds of lymphoid tissue to protect against infection

STRUCTURE & FUNCTION

- Nose:
  - Along with paranasal sinuses constitute the first part of the respiratory system
  - Responsible for receiving, filtering, warming, & moistening air to be transported to lungs
  - Also holds receptors of cranial nerve I (olfactory)
- Sinuses:
  - 4 pair: Frontal, maxillary, ethmoid, sphenoid
  - Air-filled cavities that decrease the weight of the skull & act as resonance chambers during speech
  - Lined with ciliated mucous membrane to trap debris & propel it outside

The sinuses are often a primary site of infection because they can easily become blocked. The frontal sinuses (above the eyes) and the maxillary sinuses (in the upper jaw) are accessible to examination by the nurse. The ethmoidal and sphenoidal sinuses are smaller, located deeper in the skull, and are not accessible for examination.

COLLECTING SUBJECTIVE DATA

Nose and Sinuses

- Any discharge?
- Colds, sinus pain, trauma, epistaxis, or allergies?
- Change in ability to smell or taste?

Mouth and throat

- Any sores/lesions, bleeding gums, toothache
- Dysphagia (difficulty swallowing)
- Changes in voice or hoarseness?
- Does the client smoke or drink alcohol?

Collecting subjective data related to the mouth, throat, nose and sinus can aid in detecting diseases and abnormalities and can really affect your patient's activities and daily living.
NOSE/SINUSES: Discharge can indicate chronic allergies or another serious issues like a cerebrospinal fluid leak. If you see this, it would be clear fluid and not a chronic issue and the client would have had a recent head injury.
Can you breathe through BOTH your nostrils? Inability to breath through both nostrils may indicate sinus congestion, obstruction or a deviated septum. Why do you think this is important? Effects activity, exercise, sleep.
What is epitaxis? Nosebleeds.
A decrease in the ability to smell may occur with lesions of the Optic Nerve or Facial nerve, upper respiratory tract infections, sinusitis, Alzheimer disease, Parkinson disease.

MOUTH/Throat: Red, swollen gums that bleed easily occur in early gum disease.
Dysphagia (difficulty swallowing) or odynophagia (painful swallowing) may be seen with tumors of the pharynx, esophagus, or surrounding structures, and narrowing of the esophagus such as in postradiation, gastroesophageal reflux disease (GERD), anxiety, poorly fitting dentures, or neuromuscular disorders.
A change in voice or hoarseness can be associated with upper respiratory infections,

allergies, hypothyroidism, overuse of the voice, smoking or inhaling other irritants, and cancer of the larynx.

Smoking (cigarette, pipe, cigar, smokeless tobacco) significantly increase one's risk for oropharyngeal cancer. Alcohol does as well.

OLDER ADULT QUESTIONS AND CONSIDERATIONS

ADDITIONAL QUESTIONS

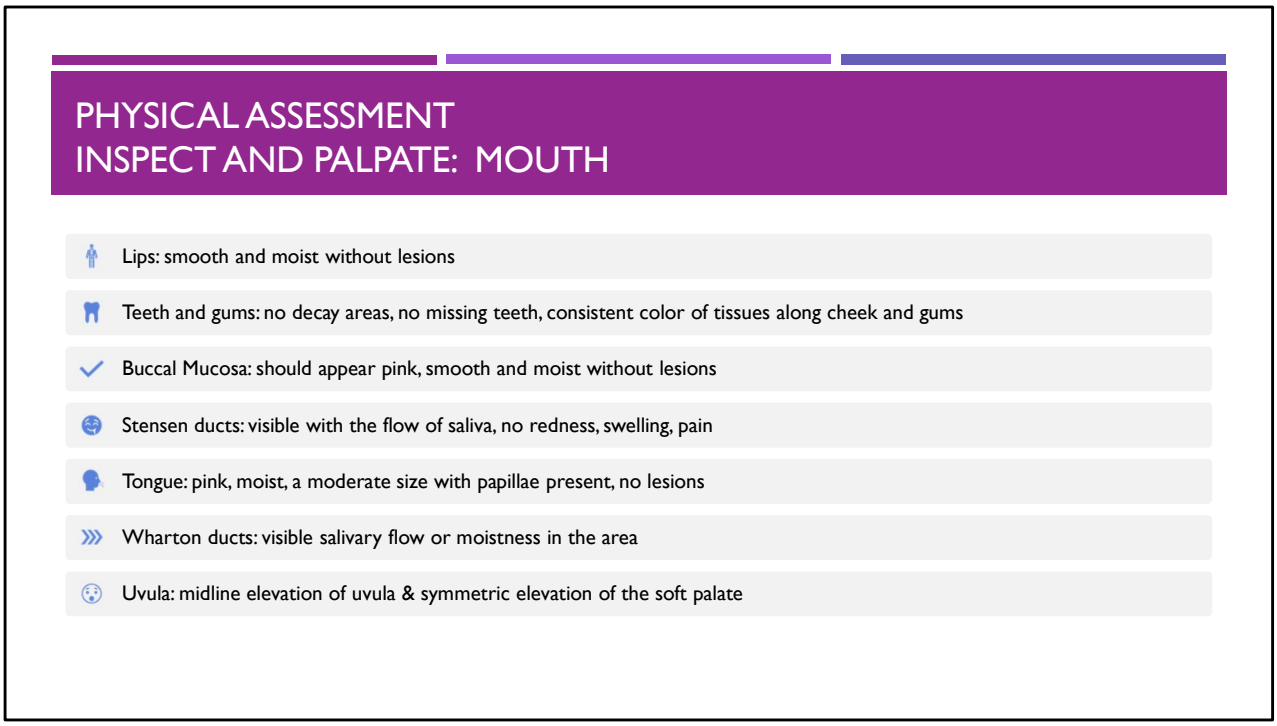
- Any dry mouth?
- Dentures, loose teeth?
- Can you chew all types of food?
- Note any change in taste or smell

CONSIDERATIONS

- Nose may appear more prominent due to loss of subcutaneous fat
- Removal of front teeth cause the mouth to look like "purse string" in appearance
- Gum margins recede thus teeth look longer; Teeth can look yellowed uniformly, due to worn dentin
  - Place the client at an increased risk for periodontal disease & tooth loss
- Tongue appears smoother due to papillary atrophy; membranes are thinned and may look shiny
- Reduction in taste and smell functioning
- Decreased salivary secretions –dysphagia (difficulty swallowing), xerostomia (dry mouth)

The oral mucosa is often drier and more fragile in the older client because the epithelial lining of the salivary glans degenerates.
Older adult teeth may appear longer because of age related gingival recession.
Parkinson's disease or Alzheimer disease can decrease the ability to smell. (decreased in sense of smell is often seen as part of the aging process)
Poor fitting dentures can lead to poor diet or mouth sores

Nose and facial bones are more prominent due to loss of subcutaneous fat
Aging adults lose the ability to identify odors.



Lips: Abnormal: pallor around the lips (anemia or shock), bluish or cyanotic lips may result from cold or hypoxia, reddish lips are seen with ketoacidosis, carbon monoxide poisoning and chronic obstructive pulmonary disease due to polycythemia.
Teeth and gums: Abnormal: missing teeth can affect chewing. Brown or yellow stains or white spots on teeth may result from antibiotic therapy or tooth trauma.

Buccal Mucosa: Abnormal: Leukoplakia (chalky white raised patches) may be seen in chronic irritation, heavy smoking, and alcohol use. These are precancerous lesions. Whitish, curd-like patches that scrape off over reddened mucosa and bleed easily indicate "thrush" candida albicans infection.

Stensen Ducts: Openings of the parotid salivary glands. Should be visible in the buccal mucosa, across from the second upper molar. Should see flow of saliva. Abnormal is reddened opening of a stensen duct is seen with mumps.

Tongue: Significant color changes of the tongue may indicate an underlying health condition. Deep longitudinal fissures seen in dehydration.
The ventral side of the tongue should be smooth, shiny, pink with visible veins and no lesions. The area underneath the tongue is the most common site of oral cancer. The

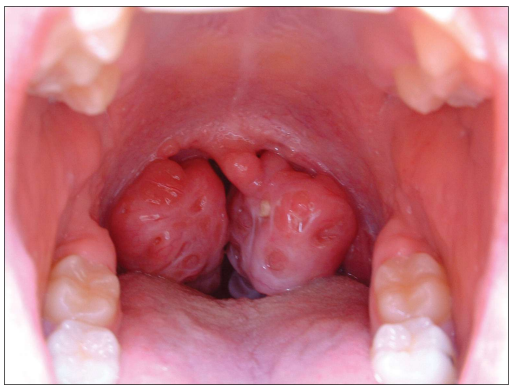
side of the tongue is the most common site of tongue cancer.

Wharton ducts: Openings from the submandibular salivary glands. On the floor of the mouth on either side of the frenulum. Abnormal findings include lesions, ulcers, nodules or hypertrophied duct openings on either side of frenulum.
Uvula: Abnormal: asymmetric movement or loss of movement may occur after a CVA (stroke).
While inspecting the mouth note any unusual or foul order: Fruity is diabetic ketoacidosis, ammonia is associated with kidney disease, foul odor may indicate an oral or respiratory infection, or tooth decay.

PHYSICAL ASSESSMENT
INSPECT AND PALPATE: MOUTH

- **Inspection of the throat and mouth: Tonsils**
  - **Grade tonsils**
  - **1+** visible
  - **2+** halfway between tonsillar pillars and uvula
  - **3+** Touching the uvula
  - **4+** Touching one another

- **Test cranial nerve XII (12-hypoglossal)** – stick out the tongue



Using the tongue depressor to keep the mouth open wide, inspect the tonsils for color, size, and presence of exudate or lesions. Shine the penlight on the back. Tonsils may be present or absent, if they had a tonsillectomy, but they should appear pink and symmetric 1+ , no exudate, swelling, or lesions.
Priority: Maintain airway

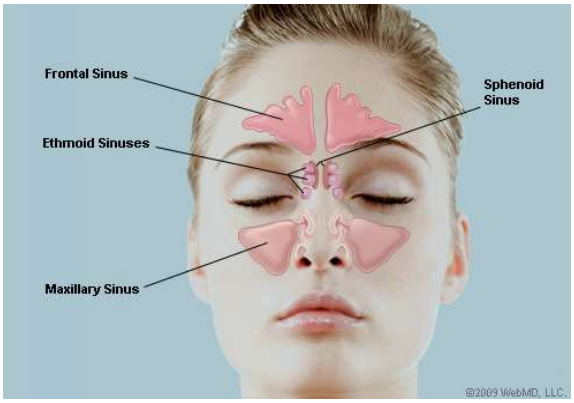
ABNORMAL FINDINGS



Koplik spots…Tiny white spots that lie over reddened mucosa. An early sign of measles.
Canker sore. Round white area with a red halo. Can occur anywhere inside mouth, on lips, tongue, buccal mucosa. Not contagious. Painful. Can be caused by stress, hormones, accidently biting yourself, food allergy…
Cold sore/fever blister. Caused by the Herpes Simplex (HSV-1) virus. Contagious until they completely crust over. Can develop inside or outside of the mouth. Can appear as one single blister, or a cluster of blisters. Stress can trigger an outbreak.

PHYSICAL ASSESSMENT
INSPECT AND PALPATE: NOSE

- **Nose: inspect and palpate**
  - Patency of the nostrils
  - Visually look at mucous membranes in the nose
  - **Palpate sinuses-** frontal and maxillary are the only two accessible to examination



Inspect & palpate the external nose: observe the nasal color, shape, consistency, and tenderness: Nasal tenderness on palpation accompanies a local infection
Patency of the nostrils: Is your client able to sniff through each nostril? Abnormal findings could indicate swelling, rhinitis, foreign object, deviated septum.

Inspecting the internal nose with an otoscope. Normal findings is dark pink, moist, and free of exudate, nasal septum is intact and free of ulcers.
Abnormal finding is the nasal mucosa is swollen and pale pink or bluish gray in clients with allergies or red and swollen with upper respiratory infection. Exudate is common with infection. Can range from watery to thick, yellow-green discharge. Purulent (thick/opaque) nasal discharge is seen with acute bacterial rhinosinusitis.

Ulcers can be (not all) with use of cocaine, trauma, chronic infection, chronic nose picking. Growths or masses on the mucosa can be seen with patient with chronic allergies… Everyone is Missouri has chronic allergies.
Trauma, foreign body, vigorous nose blowing, or HTN can all cause epistaxis (nosebleeds)

Sinus: There are four sinuses and we can only palpate 2 of them. The frontal and maxillary are the only two accessible to examination. The sphenoid

and ethmoid sinuses are two deep to palpate. Tenderness on the sinuses are abnormal and could indicate an infection.



HEALTH PROMOTION AND DISEASE PREVENTION: SINUSITIS

- 2 types:
  - Acute: Typically viral, symptoms last <3 weeks, begins with common cold and goes away within 10 days
  - Chronic: Bacterial infection, lasts >12 weeks despite medical treatment
- Causes: Infection, fungus, deviated nasal septum, nasal polyps, or immune system deficiency
- Symptoms: Thick yellow-green nasal discharge, postnasal drip, cough, toothache, fever, congestion, pain/tenderness, swelling, headache, reduced smell & taste

Individuals who suffer from chronic rhinitis or asthma are at higher risk for chronic sinusitis due to prolonged inflammation of the airways.

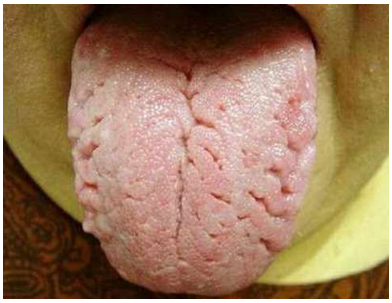
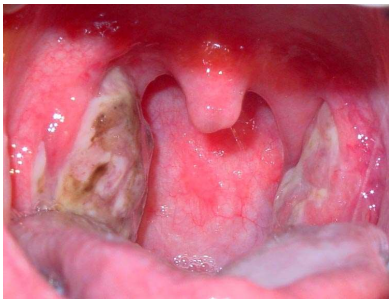
Education:
Avoid catching colds or influenza
Avoid allergy triggers
Stay hydrated
Inhale steam or rinse nose with saline solution regularly
Good hand hygiene
Get yearly influenza vaccine



HEALTHY TONGUE

Healthy tongue should be pink, moist, with many papillae, midline
No lesions
Under the tongue is the most common site of ORAL cancer
The side of the tongue is the most common site of TONGUE cancer



ABNORMAL FINDINGS

- Black hairy tongue
- Oral Candidiasis (Thrush)
- Fissured tongue
- Tonsillar abscess

Black hairy tongue-Fungal infection causes papillae to grow - due to antibiotics or Pepto Bismol, also seen with heavy smoking, alcohol intake

Fissured tongue – harmless condition
Thrush on the tongue (oral candidiasis)- Thrush can occur on the tongue or the buccal mucosa. Common in babies, immunocompromised, those with dry mouth, users of oral steroid sprays for asthma
Tonsillar abscess

ABNORMAL FINDINGS

- Smooth, beefy-red tongue without papillae from Vitamin B-12 or Niacin deficiency
- Raised, white, feathery areas: Oral hairy leukoplakia seen in HIV/AIDS or immunocompromised clients





Vitamin B12 or a Niacin deficiency can lead to swollen, beefy-red tongue with little to no papillae

HIV/AIDS or other immune compromised individuals can develop Oral hairy leukoplakia (OHL). Results in white feathery looking areas of the tongue that cannot be scraped off.

HEALTH PROMOTION AND DISEASE PREVENTION: OROPHARYNGEAL CANCER



- Can have significant effects on well-being
- Rates more than twice as high in those assigned male at birth
- Oral cancers most prevalent in the tongue, tonsils & oropharynx, gums, & floor of the mouth

Healthy People 2030 goal: Increase the percentage of oropharyngeal cancers detected at the earliest stages

The fact that the oral cavity and oropharynx, along with other parts of the head and neck, contribute to the ability to chew, swallow, breathe, and talk, oropharyngeal cancer can have significant effects on well-being.

The ACS (2020) reported that an estimated 58,000+ new cases of cancer of the oral cavity and pharynx (throat) are expected in 2024, and 12,000+ people are expected to die of these. The death rate from oral cancer is related to it usually being discovered only late in its course of development.

OROPHARYNGEAL CANCER



- Risk factors:
  - Tobacco use, heavy alcohol consumption, prolonged sun exposure, HPV virus, assigned male at birth, fair skin, age >45, poor oral hygiene, poor diet/nutrition, weakened immune system, marijuana use
- Client education:
  - Avoid smoking and tobacco use, avoid heavy alcohol use, avoid HPV (consider vaccination), avoid excessive sun exposure, eat a well-rounded diet rich in fruits, vegetables, and vitamin A, practice good oral hygiene and get routine dental care, avoid smoking marijuana especially if you have other risk factors

REVIEW QUESTION

- While in your client's room you notice his dentures are still in his case and his breakfast is barely eaten. What would be the BEST action to take?
  - A. Assess the mouth and lips for lesions or sores
  - B. Ask the patient if there is something else he would like to order for lunch
  - C. Place a consult for the dietician to see
  - D. Chart his intake and output and ask the patient if there is something wrong with the food.


Nursing school: where every answer is correct, but you're still probably wrong.
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QUESTIONS??